

24 — MKUFT Cross-Support and Traversal Map

Professional reading edition of the evolving public MKUFT canon

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Purpose

MKUFT contains several kinds of public document: foundational proposals, mathematical scaffolds, experimental programmes, falsifiers, explanatory bridges, and applied research branches. This map shows how those public objects support and constrain one another.

Its central rule is:

A module should be read with the definitions, evidence requirements, ordinary alternatives, and falsifiers that limit its claims.

The map is navigation and dependency structure. It does not create evidence, establish a physical mechanism, or replace the scientific content of the linked modules.

Reader routes

Physics and quantum gravity

README → Layer Before Law → Typed Traversal and Equation Hygiene → Context-Conditioned State Comparison → Recursive Constraint Closure → Relational Brackets / I→P Admissibility → Intrinsic-Extrinsic Transport → SIPO Capstone → Dynamic Interface Promotion → Experimental Programme → Falsification

LUCY and effective geometry

LUCY Boundary Threshold Framework → Layer Before Law → LUCY Threshold Geometry and Relational Closure → Typed Hygiene → Context-Conditioned State Comparison → Relational Brackets and I→P Admissibility → Transport / Boundary-Conditioned Realisation → SIPO Capstone → Dynamic Interface Promotion → Promotion Gates → Falsification

General architecture

Start Here → Integrated Master Spine → Canon Map → Cross-Support Map → Context-Conditioned State Comparison → Recursive Constraint Closure → Load-Bearing Relation Sets → Relational Addressability → Temporal Continuity → Relational Brackets / Completion → Intrinsic-Extrinsic Transport → SIPO Capstone → Dynamic Interface Promotion → Falsification

AI and typed knowledge architectures

ATLD Publication → Active Traversal → Strongest Fair Null → Load-Bearing Deformation → Context-Conditioned State Comparison → Recursive Constraint Closure → Transport / Readdressing → SIPO Capstone → Dynamic Interface Promotion → Typed Hygiene → Promotion Gates

Core dependency groups

Foundational architecture

Primary documents:

- [MKUFT Core Extended](#)
- [Mathematical Appendix](#)
- [Standalone Formal Addendum](#)
- [Integrated Master Spine](#)
- [SIPO Capstone](#)
- [Dynamic Interface Promotion](#)

These carry the S-I-P-O architecture, possibility and information structure, physical realisation, bounded observer terms, typed state graphs, trajectories, path-cost scaffolds, the current addressed-state → admissibility descriptor → physical-law object (D,T,W) → P propagation → P/O instrument → readdressing factorisation, and the criterion for when a load-bearing interface must become an explicit recursively updated state variable rather than remain bundled context.

Foundational physics

Primary documents:

- [Layer Before Law](#)
- [Typed Traversal and Equation Hygiene](#)
- [Context-Conditioned State Comparison and Observability](#)
- [Relational Brackets, Completion Geometry, and I→P Admissibility](#)
- [Intrinsic-Extrinsic Address Transport, Holonomy, and Boundary-Conditioned Realisation](#)
- [SIPO Capstone](#)
- [Dynamic Interface Promotion](#)

- [Physics-Facing MKUFT Explanation](#)
- [Scientific References and Current Literature](#)
- [Falsification Summary](#)

This branch carries category reframing, typed object/address discipline, admissible comparison, transport/history, dynamic interface re-entry, Bell/no-signalling requirements, quantum and gravitational recovery burdens, and the current SIPO update architecture.

The **architectural update object is now closed** by Module 33. Module 33S1 does not reopen that object; it refines the readdressing stage by specifying when a trackable interface that changes the next admissibility/law object and is itself changed by realised dynamics must re-enter the state explicitly. The branch remains incomplete as fundamental physics until the admissibility descriptor and physical-law assembler are instantiated for concrete quantum/gravitational regimes, the resulting (D,T,W) object is mathematically well-defined, accepted limits are derived, and independent predictive content is demonstrated where claimed.

Empirical discipline

Primary documents:

- [Experimental Test Programme](#)
- [Discriminating Experiments and Promotion Gates](#)
- [Falsification Summary](#)
- [Worked Examples: RNG and Environment](#)
- [Scientific Tightening and Claim Discipline](#)

These carry controls, preregistration, null models, effect-size discipline, promotion criteria, claim status, replication requirements, and reduction rules.

Observer and agency

Primary documents:

- [OCQS Human Activation Layer](#)
- [GRACE Traversal Rule](#)
- [Agency Accessibility and Capture Geometry](#)
- [Context-Conditioned State Comparison and Observability](#)
- [SIPO Capstone](#)

These carry bounded observer-state hypotheses, measurement-relative observability, action-map access, capture, responsibility, recovery, and the distinction between physical measurement back-action and O-layer record.

Boundary and physical bridges

Primary documents:

- [LUCY Boundary Threshold Framework](#)
- [LUCY Threshold Geometry and Relational Closure](#)
- [Gradient Mechanics and Boundary-Shear Geometry](#)
- [MKUFT–Dollard Field Geometry Notes](#)
- [Cross-Layer Invariants and Layer Addressing](#)
- [Context-Conditioned State Comparison and Observability](#)
- [Relational Brackets, Completion Geometry, and I→P Admissibility](#)
- [Intrinsic–Extrinsic Address Transport, Holonomy, and Boundary-Conditioned Realisation](#)
- [SIPO Capstone](#)
- [Dynamic Interface Promotion](#)
- [Typed Traversal and Equation Hygiene](#)

These carry threshold, boundary, effective-geometry hypotheses, support-conditioned comparison, completion/admissibility geometry, transport/context, law-object assembly, dynamic boundary/interface promotion, ordinary physical baselines and reduction requirements.

Procedural and Voynich applications

Primary documents:

- [ESRT / ESF and Voynich Support](#)
- [ESRT / ESF Method Appendix](#)
- [Voynich Procedural-Engine Analysis](#)

These carry addressing, state, flow, perturbation testing, false-positive controls, and the candidate procedural interpretation of the Voynich Manuscript.

Active traversal and load-bearing structure

Primary documents:

- [Active Traversal and Functional Emergence](#)
- [Strongest Fair Null and Relational Specificity](#)
- [Load-Bearing Invariants and Whole-System Deformation](#)
- [Fundamental Traversal Coherence Nodes](#)
- [Context-Conditioned State Comparison and Observability](#)
- [Recursive Constraint Closure and Reachable-State Geometry](#)
- [Load-Bearing Relation Sets and Scale-Transition Tests](#)
- [Invariant Persistence, Relational Addressability, and Scale Transition](#)

- Temporal Continuity Kernels and Minimum Identity Horizon
- Relational Brackets, Completion Geometry, and I→P Admissibility
- Intrinsic–Extrinsic Address Transport, Holonomy, and Boundary-Conditioned Realisation
- SIPO Capstone
- Dynamic Interface Promotion
- Discriminating Experiments and Promotion Gates

These carry matched architecture controls, full-history replay, relationship specificity, relation-level/node-level deformation, adaptive addressed-state comparison, local-feasibility versus higher-scale-capability tests, identity/addressability guards, temporal continuity, completion fibers, macro realisation classes, transport, recursive readdressing, explicit inter-face-state promotion, and structural versus empirical load.

Structural nodes and canonical handles

21 — Ambiguity Dynamics

Ambiguity Dynamics and Manoeuvre Space separates unresolved state volume, route connectivity, and preserved access:

$$A_{t,\text{vol}}^{(d)} = \log\left(1 + \frac{\mu_d(\Omega_t^{(d)})}{\mu_{0,d}}\right),$$

$$M_t = A_{t,\text{vol}}^{(d)} R_t X_t.$$

M_t is a heuristic audit index, not a validated universal law.

22 — Cross-Layer Invariants

Cross-Layer Invariants and Layer Addressing represents a candidate invariant at layer L as

$$K_L = A_L(K; \theta_L),$$

with typed coupling

$$C_{LM} : \mathcal{X}_L \rightarrow \mathcal{X}_M.$$

Repeated algebraic form does not establish a shared physical mechanism.

22A — Recursive Address Closure

Recursive Address Closure and Property Transmission treats an object's public identity as a typed address rather than an unqualified name. Version, evidence, licence, authority, causation, responsibility, and other properties require their own carriers. Module 33S1 applies the same recursive re-entry discipline to a dynamic interface: once a relation is both prospectively load-bearing and updated by the realised transition, it can no longer remain an unnamed piece of scenery in the next-cycle state.

23 — Agency Accessibility

Agency Accessibility and Capture Geometry uses

$$G_t(u; T_t, H_t) \in [0, 1]$$

and

$$U_t^{\text{access}} = \{u \in U_t : G_t(u; T_t, H_t) > \theta_{\text{access}}\}$$

to distinguish retained capacity from practical access under a declared model.

24A — Active Traversal

Active Traversal and Functional Emergence tests whether structured active traversal improves a predeclared functional score relative to matched controls:

$$\Delta F(q) = F[\Psi_{\text{struct}}(q)] - F[\Psi_{\text{control}}(q)].$$

24B — Strongest Fair Null

Strongest Fair Null and Relational Specificity defines the strongest fair scalar null as

$$F_{\text{null}}^* = \max\{F_A^*, F_B^*, F_{\text{ind}}^*, F_{A \rightarrow B}^*, F_{B \rightarrow A}^*, F_{\text{replay}}^*\},$$

with

$$G_{\text{syn}}^* = F_{AB} - F_{\text{null}}^*, \quad G_{\text{path}} = F_{AB} - F_{\text{replay}}^*.$$

25 — Load-Bearing Deformation

[Load-Bearing Invariants and Whole-System Deformation](#) compares baseline with controlled relation-level deformations. Structural centrality is not truth.

26 — Layer Before Law

[Layer Before Law](#) uses the typed cycle

$$S \rightarrow I \rightarrow P \rightarrow O \rightarrow S.$$

The earlier placeholder

$$U_n \xrightarrow{\text{Update}_{\text{SIPo}}} U_{n+1}$$

is now closed architecturally by Module 33 rather than remaining an unnamed task. The branch must still derive Bell/no-signalling and the accepted quantum/gravitational limits in concrete physical instantiations.

27 — Typed Traversal and Equation Hygiene

[Typed Traversal and Equation Hygiene](#) keeps component spaces typed and prevents physical geometry, state-space geometry, information adjacency, observer registration, adaptive construction address, and metaphysical language from being flattened into one mathematical object.

30 — LUCY Threshold Geometry and Relational Closure

[LUCY Threshold Geometry and Relational Closure](#) is the downstream physical application of the canonical LUCY threshold sequence. The cross-layer coupling may be zero; the branch advances only under ordinary-physics controls and accepted-limit recovery.

31 — Context-Conditioned State Comparison and Observability

[Context-Conditioned State Comparison and Observability](#) constrains counterfactuals and cross-implementation comparisons by requiring a declared comparison class, measurement-relative state equivalence, target-relative consequence spread, and support-compatible dynamics.

32 — Recursive Constraint Closure and Reachable-State Geometry

[Recursive Constraint Closure and Reachable-State Geometry](#) extends comparison discipline to adaptive systems whose active variables, state spaces, admissible actions or update rules may change.

$$\mathfrak{X} = \bigsqcup_{\alpha \in A} \{\alpha\} \times \mathcal{X}_\alpha.$$

A possible enabling-constraint signature is

$$\Delta\mu_\ell < 0, \quad \Delta\nu_L > 0,$$

where lower feasible-state volume and higher-scale capability are separately measured in lawful comparison spaces.

32S — Load-Bearing Relation Sets and Scale Transition

[Load-Bearing Relation Sets and Scale-Transition Tests](#) makes load-bearing an intervention result, distinguishes relation from carrier, and gives a promotion test for an organised lower-scale whole to earn higher-scale effective-unit status.

32S1 — Relational Addressability and Identity Parity

[Invariant Persistence, Relational Addressability, and Scale Transition](#) distinguishes physical persistence from independent relational addressability and separates within-address state equivalence, address-class parity, and cross-address continuity.

32S2 — Temporal Continuity

[Temporal Continuity Kernels and Minimum Identity Horizon](#) extends same-self comparison to recent relational trajectories where history is load-bearing.

32S3 — Relational Brackets, Completion, and Realisation Classes

[Relational Brackets, Completion Geometry, and I→P Admissibility](#) defines typed relation scopes, a compatible region

$$\mathfrak{C}_{B,t} = \{a \in \mathcal{L}_t : \chi_B(a) = 1\},$$

a missing-address completion fiber

$$\mathfrak{F}_{k,t}(a_{-k}) = \mathfrak{C}_{B,t} \cap \pi_{-k}^{-1}(a_{-k}),$$

and macrostate realisation class

$$\mathcal{M}_y = \mathfrak{C}_{B,\ell} \cap R_{\ell \rightarrow L}^{-1}(y).$$

32S4 — Intrinsic/Extrinsic Address and Transport

[Intrinsic–Extrinsic Address Transport, Holonomy, and Boundary-Conditioned Realisation](#) separates intrinsic relation class from contextual embedding, defines context-indexed P-realisation

$$\mathcal{F}_P(\kappa; b, h) = \{p \in \mathfrak{P} : \chi_P(p; \kappa, b, h) = 1\},$$

and adds lawful path/loop transport tests plus boundary-conditioned physical comparators. Same endpoint is insufficient where an actual history-dependent transport law exists.

33 — SIPO Capstone

[SIPO Capstone: Constraint-Conditioned Addressed Update Law](#) closes the architectural update object.

The complete addressed state/context first produces

$$\Xi_t = \mathcal{Q}_t(\mathcal{U}_t, E_t).$$

The active P-law object is then assembled as

$$\mathfrak{L}_{P,t} = \mathfrak{C}_P(\Xi_t, P_t, E_t) = (\mathcal{D}_{P,t}, \mathcal{T}_{P,t}, \mathcal{W}_{P,t}).$$

The compressed update is

$$\text{Update}_{\text{SIPO}} = \text{Readdress} \circ \text{Instrument}_{P \rightarrow (P,O)} \circ \text{Propagate}_{\mathfrak{L}_P^P}.$$

Domain before generator and support before weighting are retained as clean specialisations, while the general capstone permits a demonstrated relation to change domain, transition rule, weighting, or a typed combination. The P-only null remains controlling. This closes architecture, not the branch-specific quantum/gravitational derivation.

33S1 — Dynamic Interface Promotion and Recursive Boundary Closure

[Dynamic Interface Promotion and Recursive Boundary Closure](#) asks when a relation that is already permitted inside the addressed state must be made explicit as a recursively updated interface variable.

Let

$$\mathcal{J}_{AB,t}$$

be a typed interface/junction state between addressed systems A_t and B_t . Promotion requires: operational definition/trackability; a controlled prospective effect on Ξ , the active law object $(\mathcal{D}, \mathcal{T}, \mathcal{W})$, or another preregistered load-bearing outcome; and recursive back-action such that

$$\mathcal{J}_{AB,t} \rightarrow (A_{t+\Delta}, B_{t+\Delta}) \rightarrow \mathcal{J}_{AB,t+\Delta} \rightarrow \Xi_{t+\Delta}.$$

A fully P-recoverable interface can pass this state-representation gate; that does not establish independent I→P dynamics. If the coupled lower-scale object later passes the existing Module 32S promotion tests, the former boundary can become an internal relation of the promoted whole. Physical boundary work remains inside the energy ledger, and the Einstein–Rosen bridge is used only as a bounded geometric comparator rather than evidence for literal wormholes.

Complete public registry

Module	Public role	Principal limit
01 — Core Extended	S–I–P–O theory spine	Falsification / ordinary limits
02 — Mathematical Appendix	Mathematical scaffold	Operational definitions and accepted limits
03 — Formal Addendum	State graph, trajectory, cost models	Units, domains, predictive comparison
04 — Experimental Programme	Test design	Controls, preregistration, replication
05 — Falsification	Failure conditions	Null results and simpler explanations
06 — OCQS	Observer-state hypothesis	Cognitive baselines / bounded effects
07 — ESRT/ESF Support	Procedural bridge	Alternative classifications

08 – LUCY	Threshold framework	Operational thresholds / ordinary models
09 – Voynich	Candidate application	Linguistic/cipher/scrabal/decorative nulls
10 – ESRT/ESF Method	Procedural recognition method	Reliability / false positives
11 – Full-Stack Explanation	Public bridge	Canonical owners control
12 – Individual Reality Cost and Cohesion	Boundary/cohesion bridge	Analogy is not physical evidence
13 – Physics-Facing Explanation	Technical translation	Accepted-physics recovery
14 – Worked Examples	Numerical illustration	Statistics / replication
15 – Symbolic Physics	Symbolic/physics bridge	Metaphor/model/evidence separation
16 – Gradient Mechanics	Boundary-physics hypothesis	Ordinary mechanics / reproducibility
17 – Dollard Notes	Comparative bridge	Translation is not equivalence
18 – Clear Public Explanation	Plain-language synthesis	Canonical definitions control
20 – GRACE	Traversal filter	Domain evidence / falsification
21 – Ambiguity Dynamics	Manoeuvre-space model	Declared domain / simpler alternatives
22 – Cross-Layer Invariants	Typed address discipline	Layer-specific evidence / coupling tests
22A – Recursive Address Closure	Property-carrier discipline	No untyped inheritance
23 – Agency Accessibility	Action-map model	Operational rather than diagnostic use
24 – Cross-Support Map	Dependency navigation	Must reflect live canon
24A – Active Traversal	Matched-control traversal hypothesis	Matched resources / evaluation
24B – Strongest Fair Null	Fair-null/specificity protocol	Commensurable metrics / strong alternatives
25 – Load-Bearing Deformation	Relation-level deformation	Centrality is not truth
25A – Coherence Nodes	Node-level extension	Controlled substitution / reconstruction
26 – Layer Before Law	Foundational physics proposal	Bell/no-signalling / accepted-limit recovery
27 – Typed Hygiene	Mathematical/traversal discipline	Cannot replace branch-specific dynamics
28 – Promotion Gates	Evidence-tier control	Locked discriminators
29 – Claim Discipline	Claim-status discipline	Architecture/evidence/mechanism stay distinct
30 – LUCY Threshold Geometry	Downstream effective-geometry hypothesis	Conventional closure / accepted limits
31 – Context-Conditioned State Comparison	Comparison/observability discipline	Context cannot immunise claims
32 – Recursive Constraint Closure	Adaptive constraint/reachable-state geometry	Fixed-space/null alternatives
32A – Module 32 Novelty Audit	Prior-art/residual-contribution boundary	Novelty reduced by earlier equivalents
32S – Load-Bearing Relation Sets	Relation load / scale promotion	Ablation/substitution / capability tests

32S1 — Relational Addressability	Persistence/addressability/identity parity	Same substrate/name $\neq$ same-self proof
32S2 — Temporal Continuity	History-dependent continuity	History must earn load-bearing status
32S3 — Relational Brackets / I $\rightarrow$ P Admissibility	Completion, macro realisation, admissibility	Independent I $\rightarrow$ P must beat P-only null
32S4 — Intrinsic/Extrinsic Transport	Context-indexed realisation / path transport	Prior transport/boundary physics; energy accounting
33 — SIPO Capstone	Addressed admissibility and physical-law-object update factorisation	Concrete law-assembler instantiation, accepted-limit recovery and discrimination
33S1 — Dynamic Interface Promotion	Load-bearing interface re-entry and scale-role transition	Trackability, prospective discrimination, recursive back-action, P-only null and complete boundary-energy accounting

Public boundary

The public repository contains scientific proposals, mathematical and methodological material, public explanations, provenance and rights information, navigation, and deliberately published archives. This dependency map describes those public objects only and does not govern non-public development material.

Compressed rule

Define the claim at its proper layer, preserve the comparison class and construction address, build the typed admissibility descriptor, assemble the active physical law object, promote an interface into explicit state only when it is trackable, prospectively load-bearing and recursively updated, route the result to its mathematical and evidential support, keep its falsifier attached, and do not let structural importance, mathematical elegance, local restriction, relational completion, macro persistence, or local performance silently become empirical truth.